

Digital Electronics Question and Answers

Topic 2

Boolean Algebraic and Logic Gates

- 1.) Boolean Logic Operations
- 2.) Sum of Products and Products of Sum
- 3.) Karnaugh Map
- 4.) Quine McCluskey or Tabular Method of Minimization of Logic Functions
- 5.) Logic Gates and Networks – 1
- 6.) Logic Gates and Networks – 2
- 7.) Digital Integrated Circuits – 1

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Page |
2

1.) Boolean Logic Operations

1. In boolean algebra, the OR operation is performed by which properties?

- a) Associative properties
- b) Commutative properties
- c) Distributive properties
- d) All of the Mentioned

Answer: d

Explanation: The expression for Associative property is given by $A+(B+C) = (A+B)+C$ & $A*(B*C) = (A*B)*C$.

The expression for Commutative property is given by $A+B = B+A$ & $A*B = B*A$.

The expression for Distributive property is given by $A+BC=(A+B)(A+C)$ & $A(B+C) = AB+AC$.

2. The expression for Absorption law is given by _____

- a) $A + AB = A$
- b) $A + AB = B$
- c) $AB + AA' = A$
- d) $A + B = B + A$

Answer: a

Explanation: The expression for Absorption Law is given by: $A+AB = A$.

Proof: $A + AB = A(1+B) = A$ (Since $1 + B = 1$ as per 1's Property).

3. According to boolean law: $A + 1 = ?$

- a) 1
- b) A
- c) 0
- d) A'

Answer: a

Explanation: $A + 1 = 1$, as per 1's Property.

4. The involution of A is equal to _____

- a) A
- b) A'
- c) 1
- d) 0

Answer: a

Explanation: The involution of A means double inversion of A (i.e. A'') and is equal to A.

Proof: $((A)')' = A$

5. $A(A + B) = ?$

- a) AB
- b) 1
- c) $(1 + AB)$
- d) A

Answer: d

Explanation: $A(A + B) = AA + AB$ (By Distributive Property) = $A + AB$ ($A.A = A$ By Commutative Property) = $A(1 + B) = A * 1$ ($1 + B = 1$ by 1's Property) = A.

6. DeMorgan's theorem states that _____

- a) $(AB)' = A' + B'$
- b) $(A + B)' = A' * B'$
- c) $A' + B' = A'B'$
- d) $(AB)' = A' + B$

Answer: a

Explanation: The DeMorgan's law states that $(AB)' = A' + B'$ & $(A + B)' = A' * B'$, as per the Dual Property.

7. $(A + B)(A' * B') = ?$

- a) 1
- b) 0
- c) AB
- d) AB'

Answer: b

Explanation: The DeMorgan's law states that $(AB)' = A' + B'$ & $(A + B)' = A' * B'$, as per the Dual Property.

8. Complement of the expression $A'B + CD'$ is _____

- a) $(A' + B)(C' + D)$
- b) $(A + B')(C' + D)$
- c) $(A' + B)(C' + D)$
- d) $(A + B')(C + D')$

Answer: b

Explanation: $(A'B + CD')' = (A'B)'(CD')'$ (By DeMorgan's Theorem) $= (A'' + B')(C' + D'')$ (By DeMorgan's Theorem) $= (A + B')(C' + D)$.

9. Simplify $Y = AB' + (A' + B)C$.

- a) $AB' + C$
- b) $AB + AC$
- c) $A'B + AC'$
- d) $AB + A$

Answer: a

Explanation: $Y = AB' + (A' + B)C = AB' + (AB')'C = (AB' + C)(AB' + AB') = (AB' + C).1 = (AB' + C)$.

10. The boolean function $A + BC$ is a reduced form of _____

- a) $AB + BC$
- b) $(A + B)(A + C)$
- c) $A'B + AB'C$
- d) $(A + C)B$

Answer: b

Explanation: $(A + B)(A + C) = AA + AC + AB + BC = A + AC + AB + BC$ (By Commutative Property) $= A(1 + C + B) + BC = A + BC$ ($1 + B + C = 1$ By 1's Property).

2.) Sum of Products and Products of Sum

1. The logical sum of two or more logical product terms is called _____

- a) SOP
- b) POS
- c) OR operation
- d) NAND operation

Answer: a

Explanation: The logical sum of two or more logical product terms, is called SOP (i.e. sum

of product). The logical product of two or more logical sum terms is called POS (i.e. product of sums).

2. The expression $Y=AB+BC+AC$ shows the _____ operation.

- a) EX-OR
- b) SOP
- c) POS
- d) NOR

Answer: b

Explanation: The given expression has the operation product as well as the sum of that. So, it shows SOP operation. POS will be the product of sum terms.

3. The expression $Y=(A+B)(B+C)(C+A)$ shows the _____ operation.

- a) AND
- b) POS
- c) SOP
- d) NAND

Answer: b

Explanation: The given expression has the operation sum as well as the product of that. So, it shows POS(product of sum) operation. SOP will be the sum of product terms.

4. A product term containing all K variables of the function in either complemented or uncomplemented form is called a _____

- a) Minterm
- b) Maxterm
- c) Midterm
- d) Σ term

Answer: a

Explanation: A product term containing all K variables of the function in either complemented or uncomplemented form is called a minterm. A sum term containing all K variables of the function in either complemented or uncomplemented form is called a maxterm.

5. According to the property of minterm, how many combination will have value equal to 1 for K input variables?

- a) 0
- b) 1

- c) 2
- d) 3

Answer: b

Explanation: The main property of a minterm is that it possesses the value 1 for only one combination of K input variables and the remaining will have the value 0.

Page |
6

6. The canonical sum of product form of the function $y(A,B) = A + B$ is _____

- a) $AB + BB + A'A$
- b) $AB + AB' + A'B$
- c) $BA + BA' + A'B'$
- d) $AB' + A'B + A'B'$

Answer: b

Explanation: $A + B = A.1 + B.1 = A(B + B') + B(A + A') = AB + AB' + BA + BA' = AB + AB' + A'B = AB + AB' + A'B.$

7. A variable on its own or in its complemented form is known as a _____

- a) Product Term
- b) Literal
- c) Sum Term
- d) Word

Answer: b

Explanation: A literal is a single logic variable or its complement. For example — X, Y, A', Z, X' etc.

8. Maxterm is the sum of _____ of the corresponding Minterm with its literal complemented.

- a) Terms
- b) Words
- c) Numbers
- d) Nibble

Answer: a

Explanation: Maxterm is the sum of terms of the corresponding Minterm with its literal complemented.

9. Canonical form is a unique way of representing _____

- a) SOP

- b) Minterm
- c) Boolean Expressions
- d) POS

View Answer

Answer: c

Explanation: Boolean Expressions are represented through a canonical form. An example of canonical form is $A'B'C' + AB'C + ABC'$.

10. There are _____ Minterms for 3 variables (a, b, c).

- a) 0
- b) 2
- c) 8
- d) 1

Answer: c

Explanation: Minterm is given by 2^n . So, $2^3 = 8$ minterms are required.

11. _____ expressions can be implemented using either (1) 2-level AND-OR logic circuits or (2) 2-level NAND logic circuits.

- a) POS
- b) Literals
- c) SOP
- d) POS

Answer: c

Explanation: SOP expressions can be implemented using either (1) 2-level AND-OR logic circuits or (2) 2-level NAND logic circuits.

3.) Karnaugh Map

1. A Karnaugh map (K-map) is an abstract form of _____ diagram organized as a matrix of squares.

- a) Venn Diagram
- b) Cycle Diagram
- c) Block diagram
- d) Triangular Diagram

Answer: a

Explanation: A Karnaugh map (K-map) is an abstract form of Venn diagram organized as a matrix of squares, where each square represents a Maxterm or a Minterm.

2. There are _____ cells in a 4-variable K-map.

- a) 12
- b) 16
- c) 18
- d) 8

Page |
8

Answer: b

Explanation: There are $16 = (2^4)$ cells in a 4-variable K-map.

3. The K-map based Boolean reduction is based on the following Unifying Theorem: $A + A' = 1$.

- a) Impact
- b) Non Impact
- c) Force
- d) Complementarity

Answer: b

Explanation: The given expression $A + A' = 1$ is based on non-impact unifying theorem.

4. Each product term of a group, $w'.x.y'$ and $w.y$, represents the _____ in that group.

- a) Input
- b) POS
- c) Sum-of-Minterms
- d) Sum of Maxterms

Answer: c

Explanation: In a minterm, each variable w , x or y appears once either as the variable itself or as the inverse. So, the given expression satisfies the property of Sum of Minterm.

5. The prime implicant which has at least one element that is not present in any other implicant is known as _____

- a) Essential Prime Implicant
- b) Implicant
- c) Complement
- d) Prime Complement

Answer: a

Explanation: Essential prime implicants are prime implicants that cover an output of the function that no combination of other prime implicants is able to cover.

6. Product-of-Sums expressions can be implemented using _____

- a) 2-level OR-AND logic circuits
- b) 2-level NOR logic circuits
- c) 2-level XOR logic circuits
- d) Both 2-level OR-AND and NOR logic circuits

Answer: d

Explanation: Product-of-Sums expressions can be implemented using 2-level OR-AND & NOR logic circuits.

7. Each group of adjacent Minterms (group size in powers of twos) corresponds to a possible product term of the given _____

- a) Function
- b) Value
- c) Set
- d) Word

Answer: a

Explanation: Each group of adjacent Minterms (group size in powers of twos) corresponds to a possible product term of the given function.

8. Don't care conditions can be used for simplifying Boolean expressions in _____

- a) Registers
- b) Terms
- c) K-maps
- d) Latches

Answer: c

Explanation: Don't care conditions can be used for simplifying Boolean expressions in K-maps which helps in pairing with 1/0.

9. It should be kept in mind that don't care terms should be used along with the terms that are present in _____

- a) Minterms
- b) Expressions

- c) K-Map
- d) Latches

Answer: a

Explanation: It should be kept in mind that don't care terms should be used along with the terms that are present in minterms as well as maxterms which reduces the complexity of the boolean expression.

Page |
10

10. Using the transformation method you can realize any POS realization of OR-AND with only.

- a) XOR
- b) NAND
- c) AND
- d) NOR

Answer: d

Explanation: Using the transformation method we can realize any POS realization of OR-AND with only NOR.

11. There are many situations in logic design in which simplification of logic expression is possible in terms of XOR and _____ operations.

- a) X-NOR
- b) XOR
- c) NOR
- d) NAND

Answer: a

Explanation: There are many situations in logic design in which simplification of logic expression is possible in terms of XOR and XNOR operations.

Expression of XOR : $AB' + A'B$

Expression of XNOR : $AB + A'B'$

12. These logic gates are widely used in _____ design and therefore are available in IC form.

- a) Sampling
- b) Digital
- c) Analog
- d) Systems

Answer: b

Explanation: These logic gates(XOR, XNOR, NOR) are widely used in digital design and therefore are available in IC form as digital circuits deal with data transmission in the form of binary digits.

13. In case of XOR/XNOR simplification we have to look for the following _____

Page |
11

- a) Diagonal Adjacencies
- b) Offset Adjacencies
- c) Straight Adjacencies
- d) Both diagonal and offset adjencies

Answer: d

Explanation: In case of XOR/XNOR simplification we have to look for the following diagonal and offset adjacencies. XOR gives output 1 when odd number of 1s are present in input while XNOR gives output 1 when even number of 1s or all 0s are present in input.

14. Entries known as _____ mapping.

- a) Diagonal
- b) Straight
- c) K
- d) Boolean

Answer: a

Explanation: Entries known as diagonal mapping. The diagonal mapping holds true when for any relation, there is a projection of product on the factor.

4.) Quine McCluskey or Tabular Method of Minimization of Logic Functions

1. The output of an EX-NOR gate is 1. Which input combination is correct?

- a) $A = 1, B = 0$
- b) $A = 0, B = 1$
- c) $A = 0, B = 0$
- d) $A = 0, B' = 1$

Answer: c

Explanation: The output of EX-NOR gate is given by $AB + A'B'$. So, for $A = 0$ and $B = 0$ the output will be 1.

2. In which of the following gates the output is 1 if and only if at least one input is 1?

- a) AND
- b) NOR
- c) NAND
- d) OR

Answer: d

Explanation: In or gate we need at least one bit to be equal to 1 to generate the output as 1 because OR means any of the condition out of two is equal to 1 which means if at least one input is 1 then it shows output as 1.

3. The time required for a gate or inverter to change its state is called _____

- a) Rise time
- b) Decay time
- c) Propagation time
- d) Charging time

Answer: c

Explanation: The time required for a gate or inverter to change its state is called propagation time.

4. What is the minimum number of two input NAND gates used to perform the function of two input OR gates?

- a) One
- b) Two
- c) Three
- d) Four

Answer: c

Explanation: $Y = A + B$. This is the equation of OR gate. We require 3 NAND gates to create OR gate. We can also write,

1st, 2nd and 3rd NAND operations as: $Y = ((\text{NOT } A) \text{ AND } (\text{NOT } B))' = A'' + B'' = (A+B)$.

5. Odd parity of word can be conveniently tested by _____

- a) OR gate
- b) AND gate
- c) NAND gate
- d) XOR gate

Answer: d

Explanation: Odd parity of word can be conveniently tested by XOR gate, since, XOR outputs 1 only when the input has odd number of 1's.

6. The number of full and half adders are required to add 16-bit number is _____

- a) 8 half adders, 8 full adders
- b) 1 half adders, 15 full adders
- c) 16 half adders, 0 full adders
- d) 4 half adders, 12 full adders

Answer: b

Explanation: Half adder has two inputs and two outputs whereas Full Adder has 3 inputs and 2 outputs. One half adder can add the least significant bit of the two numbers whereas full adders are required to add the remaining 15 bits as they all involve adding carries.

7. Which of the following will give the sum of full adders as output?

- a) Three point major circuit
- b) Three bit parity checker
- c) Three bit comparator
- d) Three bit counter

Answer: d

Explanation: Counters are used for counting purposes in ascending or descending order. Three bit counter will give the sum of full adders as output.

8. Which of the following gate is known as coincidence detector?

- a) AND gate
- b) OR gate
- c) NOR gate
- d) NAND gate

Answer: a

Explanation: AND gate is known as coincidence detector due to multiplicity behaviour, as it outputs 1 only when all the inputs are 1.

9. An OR gate can be imagined as _____

- a) Switches connected in series
- b) Switches connected in parallel

- c) MOS transistor connected in series
- d) BJT transistor connected in series

Answer: b

Explanation: OR gate means addition of two inputs, which outputs when any of the input is high. Due to this reason, it is imagined as switches connected in parallel.

Page |
14

10. How many full adders are required to construct an m-bit parallel adder?

- a) $m/2$
- b) m
- c) $m-1$
- d) $m+1$

Answer: c

Explanation: We need adder for every bit. So we should need m bit adders. A full adder adds a carry bit to two inputs and produces an output and a carry. But the most significant bits can use a half adder which differs from the full adder as in that it has no carry input, so we need $m-1$ full adders and 1 half adder in m bit parallel adder.

5.) Logic Gates and Networks – 1

1. The output of a logic gate is 1 when all the input are at logic 0 as shown below:

INPUT		OUTPUT
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0
INPUT		OUTPUT

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

The gate is either _____

- a) A NAND or an EX-OR
- b) An OR or an EX-NOR
- c) An AND or an EX-OR
- d) A NOR or an EX-NOR

Answer: d

Explanation: The output of a logic gate is 1 when all inputs are at logic 0. The gate is NOR. The output of a logic gate is 1 when all inputs are at logic 0 or all inputs are at logic 1, then it is EX-NOR. (The truth tables for NOR and EX-NOR Gates are shown in the above table).

2. The code where all successive numbers differ from their preceding number by single bit is _____

- a) Alphanumeric Code
- b) BCD
- c) Excess 3
- d) Gray

Answer: d

Explanation: The code where all successive numbers differ from their preceding number by single bit is gray code. It is an unweighted code. The most important characteristic of this code is that only a single bit change occurs when going from one code number to next. BCD Code is one in which decimal digits are represented by a group of 4-bits each, whereas, in Excess-3 Code, the decimal numbers are incremented by 3 and then written in their BCD format.

3. The following switching functions are to be implemented using a decoder:

$$f_1 = \sum m(1, 2, 4, 8, 10, 14) \quad f_2 = \sum m(2, 5, 9, 11) \quad f_3 = \sum m(2, 4, 5, 6, 7)$$

The minimum configuration of decoder will be _____

- a) 2 to 4 line
- b) 3 to 8 line
- c) 4 to 16 line
- d) 5 to 32 line

Answer: c

Explanation: 4 to 16 line decoder as the minterms are ranging from 1 to 14.

4. How many AND gates are required to realize $Y = CD + EF + G$?

- a) 4
- b) 5
- c) 3
- d) 2

Answer: d

Explanation: To realize $Y = CD + EF + G$, two AND gates are required and two OR gates are required.

5. The NOR gate output will be high if the two inputs are _____

- a) 00
- b) 01
- c) 10
- d) 11

Answer: a

Explanation: In 01, 10 or 11 output is low if any of the I/P is high. So, the correct option will be 00.

6. How many two-input AND and OR gates are required to realize $Y = CD + EF + G$?

- a) 2, 2
- b) 2, 3
- c) 3, 3
- d) 3, 2

Answer: a

Explanation: $Y = CD + EF + G$

The number of two input AND gate = 2

The number of two input OR gate = 2.

7. A universal logic gate is one which can be used to generate any logic function. Which of the following is a universal logic gate?

- a) OR
- b) AND
- c) XOR
- d) NAND

Answer: d

Explanation: An Universal Logic Gate is one which can generate any logic function and also the three basic gates: AND, OR and NOT. Thus, NOR and NAND can generate any logic function and are thus Universal Logic Gates.

8. A full adder logic circuit will have _____

- a) Two inputs and one output
- b) Three inputs and three outputs
- c) Two inputs and two outputs
- d) Three inputs and two outputs

Answer: d

Explanation: A full adder circuit will add two bits and it will also accounts the carry input generated in the previous stage. Thus three inputs and two outputs (Sum and Carry) are there. In case of half adder circuit, there are only two inputs bits and two outputs (SUM and CARRY).

9. How many two input AND gates and two input OR gates are required to realize $Y = BD + CE + AB$?

- a) 3, 2
- b) 4, 2
- c) 1, 1
- d) 2, 3

Answer: a

Explanation: There are three product terms. So, three AND gates of two inputs are required. As only two input OR gates are available, so two OR gates are required to get the logical sum of three product terms.

10. Which of the following are known as universal gates?

- a) NAND & NOR
- b) AND & OR
- c) XOR & OR
- d) EX-NOR & XOR

Answer: a

Explanation: The NAND & NOR gates are known as universal gates because any digital circuit can be realized completely by using either of these two gates, and also they can generate the 3 basic gates AND, OR and NOT.

11. The gates required to build a half adder are _____

- a) EX-OR gate and NOR gate
- b) EX-OR gate and OR gate
- c) EX-OR gate and AND gate
- d) EX-NOR gate and AND gate

Answer: c

Explanation: The gates required to build a half adder are EX-OR gate and AND gate. EX-OR outputs the SUM of the two input bits whereas AND outputs the CARRY of the two input bits.

6.) Logic Gates and Networks – 2

1. A single transistor can be used to build which of the following digital logic gates?

- a) AND gates
- b) OR gates
- c) NOT gates
- d) NAND gates

Answer: c

Explanation: A transistor can be used as a switch. That is when base is low collector is high (input zero, output one) and base is high collector is low (input 1, output 0).

2. How many truth table entries are necessary for a four-input circuit?

- a) 4
- b) 8
- c) 12
- d) 16

Answer: d

Explanation: For 4 inputs: $2^4 = 16$ truth table entries are necessary.

3. Which input values will cause an AND logic gate to produce a HIGH output?

- a) At least one input is HIGH
- b) At least one input is LOW
- c) All inputs are HIGH
- d) All inputs are LOW

Page |
19

Answer: c

Explanation: For AND gate, the output is high only when both inputs are high. That's why the high output in AND will occur only when all the inputs are high. However, in case of OR gate, if at least one input is high, the output will be high.

4. Exclusive-OR (XOR) logic gates can be constructed from what other logic gates?

- a) OR gates only
- b) AND gates and NOT gates
- c) AND gates, OR gates, and NOT gates
- d) OR gates and NOT gates

Answer: c

Explanation: Expression for XOR is: $A.(B') + (A').B$

So in the above expression, the following logic gates are used: AND, OR, NOT.

Thus, 2 AND gates with two-inputs and 1 OR gate with two-inputs will be required for constructing a XOR gate.

5. The basic logic gate whose output is the complement of the input is the _____

- a) OR gate
- b) AND gate
- c) INVERTER gate
- d) XOR gate

View Answer

Answer: c

Explanation: It is also called NOT gate and it simply inverts the input, such that 1 becomes 0 and 0 becomes 1.

6. The AND function can be used to _____ and the OR function can be used to _____

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- a) Enable, disable
- b) Disable, enable
- c) Synchronize, energize
- d) Detect, invert

Answer: a

Explanation: The AND gate and OR gate are used for enabling and disabling respectively because of their multiplicity and additivity property. The AND gate outputs 1 when all inputs are at logic 1, whereas the OR gate outputs 0 when all inputs are at logic 0.

7. The dependency notation " ≥ 1 " inside a block stands for which operation?

- a) OR
- b) XOR
- c) AND
- d) XNOR

Answer: a

Explanation: The dependency notation " ≥ 1 " inside a block stands for OR operation.

8. If we use an AND gate to inhibit a signal from passing one of the inputs must be _____

- a) LOW
- b) HIGH
- c) Inverted
- d) Floating

Answer: a

Explanation: AND gate means $A \cdot B$ and OR gate means $A + B$ and to inhibit means to get low signal, one of the input must be low. It means ($0 \cdot 1 = 0$ or $1 \cdot 0 = 0$) we will get low output signal. Thus, AND gate outputs 1 only when all inputs are at logic level 1 else it outputs 0.

9. Logic gate circuits contain predictable gate functions that open their _____

- a) Outputs
- b) Inputs
- c) Pre-state
- d) Impedance state

Answer: b

Explanation: Logic gate circuits contain predictable gate functions that open their inputs because we are free to give any types of inputs.

10. How many NAND circuits are contained in a 7400 NAND IC?

- a) 1
- b) 2
- c) 4
- d) 8

Answer: c

Explanation: 7400 IC's pin has total 14 pin. Pin no 7 use for GND and pin no 14 used for +vcc and remaining pins used for connections. For a NAND gate two inputs are required and one output is obtained means for NAND gate 3 pin connections are required. Thus, a 7400IC contains 4 NAND gates with each having 3 pins. Therefore, total 12 pins dedicated for the NAND operation. Rest 2 pins for power supply.

7.) Digital Integrated Circuits

1. Which of the following logic families has the highest maximum clock frequency?

- a) S-TTL
- b) AS-TTL
- c) HS-TTL
- d) HCMOS

Answer: b

Explanation: AS-TTL (Advanced Schottky) has a maximum clock frequency of 105 MHz. S-TTL (Schottky High Speed TTL) has 100 MHz. Found nothing as HS-TTL. There are H and S separate TTL. HCMOS has 50 MHz clock frequency.

2. Why is the fan-out of CMOS gates frequency dependent?

- a) Each CMOS input gate has a specific propagation time and this limits the number of different gates that can be connected to the output of a CMOS gate
- b) When the frequency reaches the critical value the gate will only be capable of delivering 70% of the normal output voltage and consequently the output power will be one-half of normal and this defines the upper operating frequency
- c) The higher number of gates attached to the output the more frequently they will have to be serviced thus reducing the frequency at which each will be serviced with an input signal
- d) The input gates of the FETs are predominantly capacitive and as the signal frequency increases the capacitive loading also increases thereby limiting the number of loads that may be attached to the output of the driving gate

Answer: d

Explanation: Fan out is the measure of maximum number of inputs that a single logic gate output can drive. Actually power dissipation in CMOS circuits depends on clock frequency. As the frequency increases P_d also increases so fan-out depends on frequency.

3. Logic circuits that are designated as buffers, drivers or buffers/drivers are designed to have _____

- a) A greater current/voltage capability than an ordinary logic circuit
- b) Greater input current/voltage capability than an ordinary logic circuit
- c) A smaller output current/voltage capability than an ordinary logic
- d) Greater the input and output current/voltage capability than an ordinary logic circuit

Answer: a

Explanation: Buffer circuits are usually incorporated to isolate the input from the output. Logic circuits that are designated as buffers, drivers or buffer/drivers are designed to have a greater current/voltage capability than an ordinary logic circuit

4. Which of the following will not normally be found on a data sheet?

- a) Minimum HIGH level output voltage
- b) Maximum LOW level output voltage
- c) Minimum LOW level output voltage
- d) Maximum HIGH level input current

Answer: c

Explanation: Minimum LOW level output voltage will not normally be found on a data sheet.

5. Which of the following logic families has the shortest propagation delay?

- a) S-TTL
- b) AS-TTL
- c) HS-TTL
- d) HCMOS

Answer: b

Explanation: AS-TTL (Advanced Schottky) has a maximum clock frequency that is 105 MHz. So, the propagation delay will be given by $1/105$ sec which is the lowest one. It is followed by S-TTL and HCMOS in terms of increasing propagation delay.

6. What is the static charge that can be stored by your body as you walk across a carpet?

- a) 300 volts
- b) 3000 volts

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- c) 30000 volts
- d) Over 30000 volts

Answer: d

Explanation: When a person walks across a carpeted or tile floor electric charge builds up in the body due to the friction between shoes and floor material. If the friction static is greater the voltage potential develop in the body will be greater. You start act as a capacitor. This is called Electrostatic discharge. The potential static charge that can develop from walking on tile floors is greater than 15000 volts while carpeted floors can generate in excess of 30000 volts.

Page |
23

7. What must be done to interface TTL to CMOS?

- a) A dropping resistor must be used on the CMOS of 12 V supply to reduce it to 5 V for the TTL
- b) As long as the CMOS supply voltage is 5 V they can be interfaced (however, the fan-out of the TTL is limited to five CMOS gates)
- c) A 5 V zener diode must be placed across the inputs of the TTL gates in order to protect them from the higher output voltages of the CMOS gates
- d) A pull-up resistor must be used between the TTL output-CMOS input node and Vcc; the value of RP will depend on the number of CMOS gates connected to the node

Answer: d

Explanation: To interface TTL to CMOS a pull-up resistor must be used between the TTL output-CMOS input node and Vcc. A pull-up resistor is used to avoid the floating state on the input node of the CMOS, thus using a small amount of current. The value of RP will depend on the number of CMOS gates connected to the node.

8. What causes low-power Schottky TTL to use less power than the 74XX series TTL?

- a) The Schottky-clamped transistor
- b) A larger value resistor
- c) The Schottky-clamped MOSFET
- d) A small value resistor

Answer: b

Explanation: A larger value resistor causes low power low-power Schottky TTL to use less power than the 74XX series TTL.

9. What are the major differences between the 5400 and 7400 series of ICs?

- a) The 5400 series are military grade and require tighter supply voltages and temperatures
- b) The 5400 series are military grade and allow for a wider range of supply voltages and

temperatures

c) The 7400 series are an improvement over the original 5400s

d) The 7400 series was originally developed by Texas Instruments and the 5400 series was brought out by National Semiconductors after TI's patents expired as a second supply source

Answer: b

Explanation: The 5400 series are military grade and allow for a wider range of supply voltages and temperatures, these are the major differences between the 5400 and 7400 series of ICs. Also, the working temperature range of 5400 series is -50 to 125C while that for 7400 is 0 to 70C.

10. Which of the following statements apply to CMOS devices?

a) The devices should not be inserted into circuits with the power on

b) All tools, test equipment and metal workbenches should be tied to earth ground

c) The devices should be stored and shipped in antistatic tubes or conductive foam

d) All of the Mentioned

Answer: d

Explanation: For CMOS devices, all the mentioned statements are applicable. The devices should not be inserted into circuits with the power on. All tools, test equipment and metal workbenches should be tied to earth ground. Also, the devices should be stored and shipped in antistatic tubes or conductive foam.

11. SSI refers to _____

a) Small Scale Integration

b) Short Scale Integration

c) Small Set Integration

d) Short Set Integration

Answer: a

Explanation: SSI refers to Small Scale Integration.

12. Small Scale Integration (SSI) refers to ICs with _____ gates on the same chip.

a) Fewer than 10

b) Greater than 10

c) Equal to 10

d) Greater than 50

Answer: a

Explanation: Small Scale Integration (SSI) refers to ICs with fewer than 10 gates on the same chip.

13. MSI means _____
- a) Merged Scale Integration
 - b) Main Scale Integration
 - c) Medium Scale Integration
 - d) Main Set Integration

Answer: c

Explanation: MSI means Medium Scale Integration.

14. MSI includes _____ gates per chip.
- a) 12 to 100
 - b) 13 to 50
 - c) greater than 10
 - d) greater than 100

Answer: a

Explanation: Medium Scale Integration includes 12 to 100 gates per chip.

15. LSI means _____ and refers to _____ gates per chip.
- a) Long Scale Integration, more than 10 upto 10000
 - b) Large Scale Integration, more than 100 upto 5000
 - c) Large Short Integration, less than 10 and greater than 5000
 - d) Long Short Integration, more than 10 upto 10000

Answer: b

Explanation: The full form of LSI is Large Scale Integration and refers to more than 100 upto 5000 gates per chip.

16. Integrated circuits are classified as _____
- a) Large, Small and Medium
 - b) Very Large, Small and Linear
 - c) Linear and Digital
 - d) Non-Linear and Digital

Answer: c

Explanation: Integrated circuits are classified as Linear and Digital. Linear operates with continuous and digital refers to discrete signals.

17. According to the IC fabrication process logic families can be divided into two broad categories as _____

- a) RTL and TTL
- b) HTL and MOS
- c) ECL and DTL
- d) Bipolar and MOS

Answer: d

Explanation: According to the IC fabrication process logic families can be divided into two broad categories as: Bipolar and Metal-oxide semiconductor. The mentioned all others are part of bipolar. Bipolar IC fabrication refers to TTL logic where MOS refers to CMOS logic.

18. The full form of DIP is _____

- a) Dual-in-Long Package
- b) Dual-in-Line Package
- c) Double Integrated Package
- d) Double-in-Line Package

Answer: b

Explanation: The full form of DIP is Dual-in-Line Package.

19. LCC refers to _____

- a) Longest Chip Carrier
- b) Leadless Chip Carrier
- c) Leaded Chip Carrier
- d) Large Chip Carrier

Answer: b

Explanation: LCC refers to Leadless Chip Carrier.

20. PGA refers to _____

- a) Plastic Grid Array
- b) Pin Grid Array
- c) Pin Greater Array
- d) Plastic Greater Array

Answer: b

Explanation: PGA refers to Pin Grid Array.

21. MOS families includes _____

- a) PMOS and NMOS

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- b) CMOS and NMOS
- c) PMOS, NMOS and CMOS
- d) EMOS, NMOS and PMOS

Answer: c

Explanation: Metal Oxide Semiconductor families includes PMOS, NMOS and CMOS.

Page |
27

22. CMOS refers to _____

- a) Continuous Metal Oxide Semiconductor
- b) Complementary Metal Oxide Semiconductor
- c) Centred Metal Oxide Semiconductor
- d) Concrete Metal Oxide Semiconductor

Answer: b

Explanation: CMOS refers to Complementary Metal Oxide Semiconductor.

23. Propagation delay is defined as _____

- a) the time taken for the output of a gate to change after the inputs have changed
- b) the time taken for the input of a gate to change after the outputs have changed
- c) the time taken for the input of a gate to change after the intermediates have changed
- d) the time taken for the output of a gate to change after the intermediates have changed

Answer: a

Explanation: Propagation delay is defined as the time taken for the output of a gate to change after the inputs have changed.

24. Propagation delay times can be divided as _____

- a) $t(\text{PLH})$ and $t(\text{LPH})$
- b) $t(\text{LPH})$ and $t(\text{PHL})$
- c) $t(\text{PLH})$ and $t(\text{PHL})$
- d) $t(\text{HPL})$ and $t(\text{LPH})$

Answer: c

Explanation: Propagation delay times can be divided as: $t(\text{PLH})$ and $t(\text{PHL})$. $t(\text{PLH})$ stands for propagation from low to high and $t(\text{PHL})$ stands for propagation from high to low.

25. The delay times are measured between the _____ % voltage levels of the input and output waveforms.

- a) 50
- b) 75
- c) 25

d) 100

Answer: a

Explanation: Propagation delay is the time taken by the output to change its state when the input changes. The average of the two propagation delays is given by $(t_1 + t_2)/2$, which gives the intermediate value. So, the delay times are measured between the 50% voltage levels of the input and output waveforms.

Page |
28

26. Power Dissipation in DIC is expressed in _____

- a) Watts or kilowatts
- b) Milliwatts or nanowatts
- c) DB
- d) Mdb

Answer: b

Explanation: Power Dissipation in DIC is expressed in milliwatts or nanowatts.

27. Fan-in is defined as _____

- a) the number of outputs connected to gate without any degradation in the voltage levels
- b) the number of inputs connected to gate without any degradation in the voltage levels
- c) the number of outputs connected to gate with degradation in the voltage levels
- d) the number of inputs connected to gate with degradation in the voltage levels

Answer: b

Explanation: Fan-in is defined as the maximum number of inputs that can be connected to the output of a gate without any degradation in the voltage levels. For example, an eight-input gate requires one Unit Load per input. Its fan-in is 8.

28. The maximum noise voltage that may appear at the input of a logic gate without changing the logical state of its output is termed as _____

- a) Noise Margin
- b) Noise Immunity
- c) White Noise
- d) Signal to Noise Ratio

Answer: b

Explanation: The maximum noise voltage that may appear at the input of a logic gate without changing the logical state of its output is termed as noise immunity.

29. Depending upon the flow of current from the output of one logic circuit to the input of another the logic families can be divided into _____ categories.

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- a) 2
- b) 3
- c) 4
- d) 5

Answer: a

Explanation: Depending upon the flow of current from the output of one logic circuit to the input of another the logic families can be divided into two categories and they are current sourcing and current sinking. Source means from where the current originates or exits while Sink means where the current enters or is accepted.

Page |
29

30. Fan-in and Fan-out are the characteristics of _____

- a) Registers
- b) Logic families
- c) Sequential Circuits
- d) Combinational Circuits

Answer: b

Explanation: Fan-in and Fan-out are the characteristics of logic families. Fan-in is the measure of maximum number of inputs that a single gate output can drive or accept. Whereas, Fan-out means the maximum number of inputs that can be fed by a single output.

1) Write the POS form for a 3-input XNOR gate? Is it canonical?

Answer:

$$Y = A \text{ XNOR } B \text{ XNOR } C$$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$$Y = (A'B'C + A'BC' + AB'C' + ABC)'$$

$$\text{So Y is POS form is } (A+B+C')(A+B'+C)(A'+B+C)(A'+B'+C')$$

As each term in the POS form is maxterm, this is canonical.

2. Why do we write 00 01 11 10 in that order while Drawing K-maps?

Answer:

In K-Map, the Boolean simplification is done by grouping the adjacent cells that have 1. To get the simplified expression, the adjacent cells must have 1 bit change. So gray code is used.

3. How many cells will a n-input variable have in K-Map?

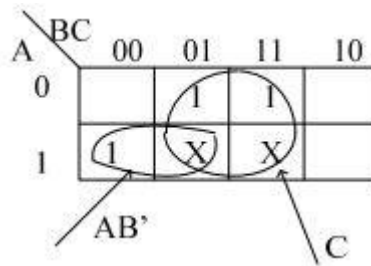
Answer:

2^n . E.g.: 3 variables, 8 cells. Similarly..4 variables 16 cells.

4. $Y = A'C + AC'B'$ and you are given that $A=C=1$ will never occur. Simplify Y?

Answer:

$Y = A'C + AC'B'$ and the output will be don't care for $A = C = 1$. So the K-map will be as follows:



Thus the simplified expression for Y is $AB' + C$

5. If $F(A,B,C,D,E) = B'E$, how many terms will be there in the standard or canonical SOP representation of F?

Answer:

8 terms, $F = B'E (A + A') (C + C') (D + D')$

Q14) In a 6 variable K-map, how many literals will the grouping of 4 adjacent cells give in the term

Answer:

$$6 - \log_2 4 = 6 - 2 = 4$$

6. In the simplification of a Boolean function, $F = \sum (0,1,2,6,7,8,9,10,14,15)$ using QM method the following table is obtained: Q1) Define: SOP form and POS form?

	0	1	2	6	7	8	9	10	14	15
BC				X	X				X	X
CD'			X	X				X	X	
B'C'	X	X				X	X			
B'D'	X		X			X		X		

- Identify the essential prime Implicants?
- Find out the simplified expression for F in SOP form?

Answer:

Checkout for the columns which has only one entry (X), that term must be included in the simplified expression. So, that term will be essential.

(a) So the essential prime implicants are: BC and $B'C'$

(b) The simplified expression $F = BC + B'C' + CD' = BC + B'C' + B'D'$

